

SpaceCDF Mission Design Course

*Collaborative Space Mission Design: From Problem to
Flight-Ready CubeSat — 40-Hour Programme*

SpaceCDF · 40 hours

Summer 2026

Course title

Collaborative Space Mission Design: From Problem to Flight-Ready CubeSat. A 40-hour intensive in concurrent space-mission design, delivered with the SpaceCDF AI-assisted Concurrent Design Facility.

Audience

Engineers, scientists, and project managers participating in concurrent design facility (CDF) studies. **No prior spacecraft design experience is required**, but a working engineering background is assumed (mechanics, electromagnetics, basic programming).

Structure

The 40-hour programme runs over **five days × eight hours** or **ten half-days**. The cohort works in **interdisciplinary teams of 4–5**, with each student taking one CDF position (e.g. Power, AOCS, Comms, Payload). Two reference texts accompany the course:

- **The Facilitator’s Book** — complete teaching reference with every session expanded, solutions, and diagrams.
- **The Learner’s Workbook** — condensed content plus worksheets for the SpaceCDF tool exercises.

EXPECTED READING

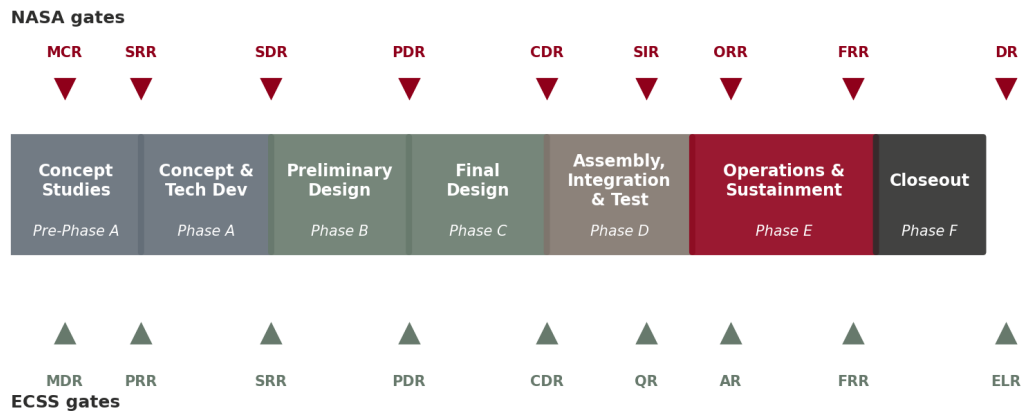
Expected reading NASA CubeSat 101 (≈ 90 min) — https://www.nasa.gov/wp-content/uploads/2017/03/nasa_csli_cubesat_101_508.pdf. NASA Systems Engineering Handbook §1 – §3 (≈ 90 min) — <https://www.nasa.gov/reference/systems-engineering-handbook/>.

What the cohort will build

Each team will, by the end of the week, deliver:

1. **A complete preliminary CubeSat mission design** — concept, requirements baseline, all major subsystem decisions, interface matrix, V&V plan, cost estimate, and risk register.
2. **An ECSS-aligned document set** auto-exported by SpaceCDF: MRD (mission-requirements document), TS (technical specification), VP (verification plan).
3. **A PDR-style design review presentation** showing budget closure, traded options, and the path to detailed design.

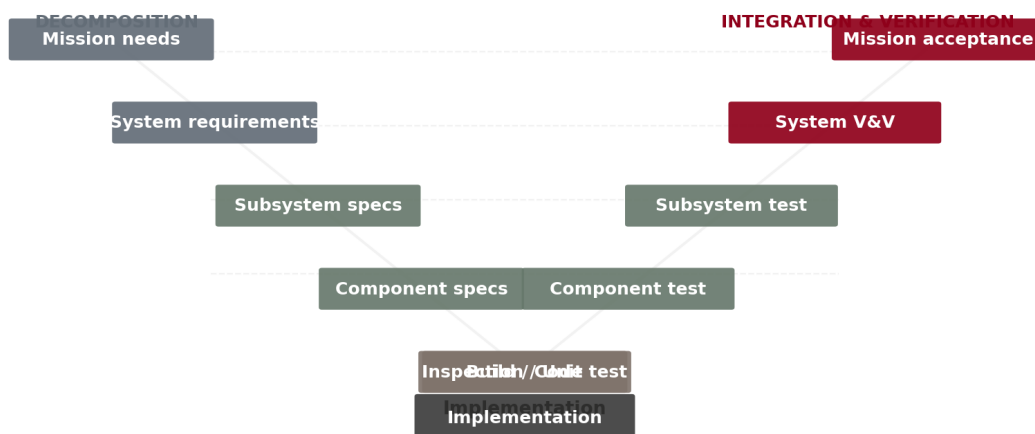
Project Lifecycle Phases & Review Gates — NASA vs ECSS



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Figure 2.1 — The lifecycle the course works inside. The 40-hour programme spans Pre-Phase A through end of Phase B (PDR).

System-V Model — Decomposition · Integration



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Figure 2.2 — System-V model. Decomposition (left) drives the build; integration and verification (right) closes the loop. The horizontal dashed lines are the traceability that ECSS and NASA both demand.

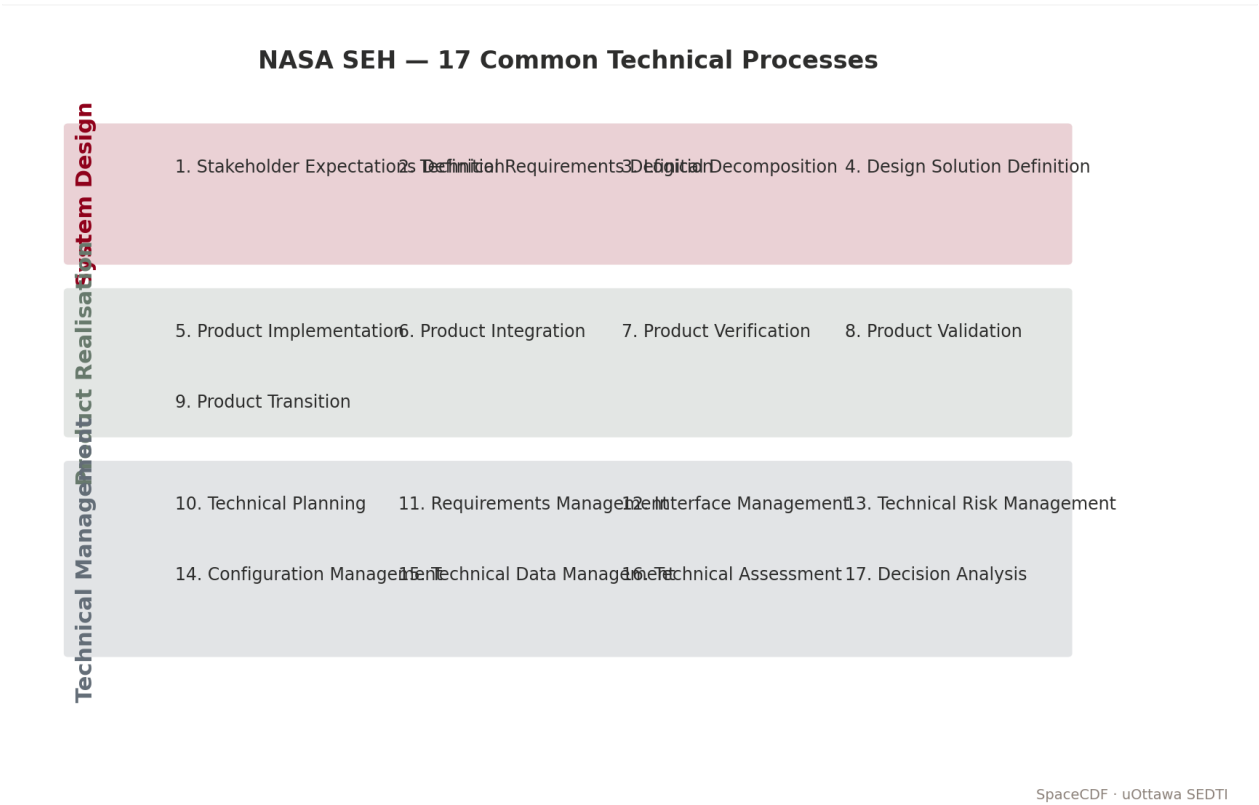


Figure 2.3 — The 17 NASA SEH processes, organised into System Design (top), Product Realisation (middle), and Technical Management (bottom). Each course session names which process(es) it sits inside.

Reference standards register

Standard	Title	Where in this course
NASA SP-2016-6105 Rev 2	Systems Engineering Handbook	All sessions; primary text for SE process
NPR 7123.1D	NASA Systems Engineering Processes & Requirements	Sessions 1.1, 2.1, 4.1, 5.1
NPR 7120.5F	NASA Space Flight Program & Project Management	Sessions 1.1, 4.4, 5.1
ECSS-E-ST-10C Rev. 1	Space engineering — general requirements	All sessions
ECSS-E-ST-10-02C Rev. 1	Verification	Sessions 4.1, 4.2
ECSS-E-ST-10-24C	Interface management	Session 2.3
ECSS-M-ST-10C Rev. 1	Project planning & management	Sessions 1.1, 4.4
ECSS-M-ST-80C	Risk management	Session 4.3
ECSS-Q-ST-30-02C	FMEA / FMECA	Session 4.3
ECSS-E-ST-32C	Structures	Session 3.2
ECSS-E-ST-31C	Thermal control	Session 3.3
ECSS-E-ST-50-05C	Radio frequency & modulation	Session 3.4
ECSS-U-AS-10C Rev. 2	Space debris mitigation	Session 5.2
Cal Poly CDS Rev 14	CubeSat Design Specification	Sessions 3.2, 5.3
CCSDS PUS	Packet Utilization Standard	Session 4.1, Week 3
ITU Radio Regulations	RF allocation & licensing	Session 5.2
ISED CPC-2-6-02	Spectrum licensing of space stations (Canada)	Session 5.2
RSSSA	Remote Sensing Space Systems Act (Canada)	Session 5.2

STANDARD REFERENCE

Standard reference ECSS standards are freely available at <https://ecss.nl/>. NASA standards: <https://standards.nasa.gov>.

Module map — sessions, NASA SEH §, ECSS §, SpaceCDF screens

Day	Session	NASA SEH §	ECSS §	SpaceCDF tab	Deliverable
1	1.1 Introduction & life-cycle	§1 – §3	E-ST-10C §4	Step 1	Tailoring matrix
1	1.2 Mission need & stakeholders	§4	E-ST-10C §5.2	Step 1	Stakeholder matrix, objectives
1	1.3 Mission trade analysis	§6.5 (Process 17)	E-ST-10C §5.3	Step 2	Trade study matrix
1	1.4 Concept of operations	App. S	E-ST-10C §5.4	ConOps tab	ConOps outline
2	2.1 Requirements engineering	§4.2	E-ST-10C §5.5	Requirements tab	Requirements baseline
2	2.2 Functions & decomposition	§4.3	E-ST-10C §5.6	Functions tab	Functional baseline
2	2.3 Interface management	§6.3	E-ST-10-24C	Interfaces tab	N ² matrix
2	2.4 Architecture & block diagrams	§4.4	E-ST-10C §5.7	Dashboard	Block diagram
3	3.1 Power & EPS	§4.4	E-ST-20C	EPS tab	Solar/battery sizing
3	3.2 Structure & mechanisms	§4.4	E-ST-32C	Structure tab	Mass budget
3	3.3 Thermal	§4.4	E-ST-31C	Thermal tab	Thermal budget
3	3.4 Comms	§4.4	E-ST-50-05C	Link Budget tab	Link budget
4	4.1 Equipment selection	§6.5	E-ST-10C §5.8	Equipment tab	BOM
4	4.2 V&V planning	§6.4	E-ST-10-02C	V&V Matrix	Verification matrix
4	4.3 Risk management	§6.6	M-ST-80C	Risk tab	Risk register
4	4.4 Cost & schedule	§6.7	M-ST-10C	Cost tab	Cost estimate
5	5.1 Gate review preparation	§7	M-ST-10C §6	Gate Review	PDR pack
5	5.2 Regulatory & licensing	—	(national)	Regulatory	Filings package
5	5.3 Launch integration	§6.5	E-ST-10C §5.10	Launch ICD	ICD checklist
5	5.4 Optimisation & final review	§6.5	E-ST-10C §5.11	Optimizer	Final design

Course Gantt — 40-hour and 3-week intensive variants

The same content can be delivered as either a 5-day 40-hour intensive (one session per slot) or a 3-week version with spreading (see the 3-week syllabus). The session sequence does not change.

	Mon	Tue	Wed	Thu	Fri
AM	1.1 + 1.2	2.1 + 2.2	3.1 + 3.2	4.1 + 4.2	5.1 + 5.2
PM	1.3 + 1.4	2.3 + 2.4	3.3 + 3.4	4.3 + 4.4	5.3 + 5.4

Day 1 — Mission Definition & Concept (8 h)

SESSION 1.1 — INTRODUCTION TO SPACE MISSION DESIGN (2 H)

- What is a Concurrent Design Facility (CDF)? History (ESA Concurrent Design Facility, JPL Team-X, NASA Glenn COMPASS).
- The System-V model (NASA SEH Ch. 2 – 3) — see Figure 2.2.
- The 17 SE processes (NPR 7123.1) — see Figure 2.3.
- NASA / ECSS lifecycle phases and review gates — see Figure 2.1.
- Role of each CDF position and what “ownership” means in a CDF.
- **Exercise:** map the 17 processes to the SpaceCDF tool tabs.
- **Worksheet:** Worksheet 1.1 (process map).

SESSION 1.2 — MISSION NEED & STAKEHOLDER ANALYSIS (2 H)

- NASA SEH Process 1: Stakeholder Expectations Definition.
- Problem-statement writing (NASA SEH §4.1).
- Stakeholder identification and needs elicitation. Indigenous community engagement and UNDRIP considerations for Canadian context.
- Objective definition with measurable success criteria.
- MoE / MoP / TPM hierarchy.
- **Exercise:** define mission need for a sample problem (using SpaceCDF Step 1).
- **Worksheet:** stakeholder matrix, objective hierarchy.

SESSION 1.3 — MISSION TRADE ANALYSIS (2 H)

- NASA SEH Process 17: Decision Analysis.
- Space vs non-space alternatives — when not to build a satellite.
- Trade-study methodology: criteria, weightings, scoring; AHP and weighted-sum sensitivity.
- Existing data services (Copernicus / Sentinel, Planet, Spire, Radarsat).
- **Exercise:** run mission trade in SpaceCDF (Step 2).
- **Worksheet:** trade-study matrix with scoring.

SESSION 1.4 — CONCEPT OF OPERATIONS (2 H)

- NASA SEH Appendix S: ConOps structure.
- Mission architecture: space, ground, user segments.
- Mission phases (LEOP → commissioning → nominal → disposal) — see Figure 2.4.
- Operational modes and duty cycling.
- Data-flow pipeline design.
- Ground-segment architecture options.
- **Exercise:** build ConOps in SpaceCDF.
- **Worksheet:** ConOps outline per Appendix S.

Mission Operations Timeline (representative LEO mission)

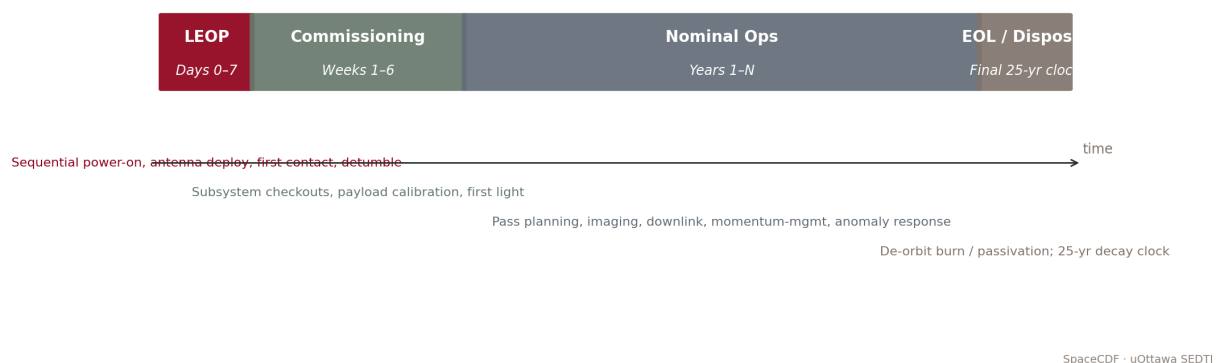


Figure 2.4 — Representative LEO mission timeline. The course designs from Day 1 against this timeline.

EXPECTED READING

Expected reading SMAD4 Ch. 4 – §6 (mission definition); NASA SEH Appendix C (How to write a good requirement) — <https://www.nasa.gov/reference/systems-engineering-hand-book/>.

Day 2 — Requirements, functions & interfaces (8 h)

SESSION 2.1 — REQUIREMENTS ENGINEERING (2 H)

- NASA SEH Process 2: Technical Requirements Definition.
- SMART requirements (Specific · Measurable · Achievable · Relevant · Traceable).
- WHAT not HOW — requirements vs design choices. The classic failure mode: writing implementation in the requirements.
- Requirement hierarchy: mission → system → subsystem.
- NASA SEH Appendix C — How to write a good requirement.
- **Exercise:** generate requirements from objectives in SpaceCDF.
- **Worksheet:** requirements quality checklist.

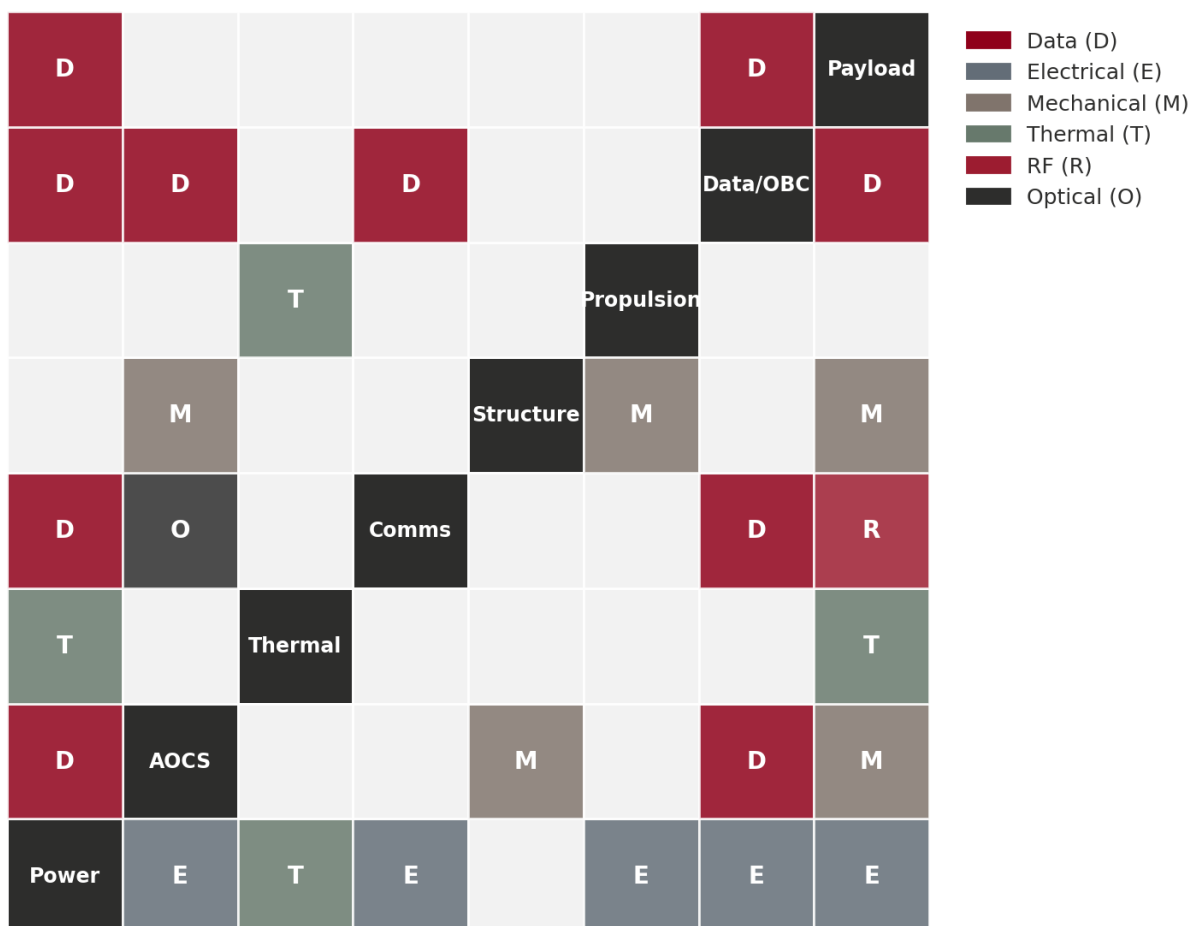
SESSION 2.2 — FUNCTIONS & DECOMPOSITION (2 H)

- NASA SEH Process 3: Logical Decomposition.
- Functional analysis vs physical decomposition.
- Function trees; function-allocation matrix; mode matrix.
- **Exercise:** decompose top-level mission function in SpaceCDF.
- **Worksheet:** function-allocation matrix.

SESSION 2.3 — INTERFACE MANAGEMENT (2 H)

- NASA SEH §6.3 (Process 12) — interface management.
- ECSS-E-ST-10-24C — interface requirements documents.
- The N^2 matrix and interface-type classification (Mechanical, Electrical, Thermal, Data, RF, Optical).

N^2 Interface Matrix — 6U CubeSat (illustrative)



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Figure 2.5 — N^2 interface matrix for a 6U CubeSat. Cells are coloured by interface type; “D” = data, “E” = electrical, “M” = mechanical, “T” = thermal, “R” = RF, “O” = optical.

- **Exercise:** complete N^2 matrix for the team’s mission in SpaceCDF.

- **Worksheet:** interface-requirements draft for two pairs.

SESSION 2.4 — ARCHITECTURE & BLOCK DIAGRAMS (2 H)

- Architecture decision records.
- Block-diagram conventions; signal-flow vs power-flow.
- Power architecture; data architecture; thermal coupling.
- **Exercise:** build subsystem block diagrams in SpaceCDF.
- **Worksheet:** architecture decision log (3 entries).

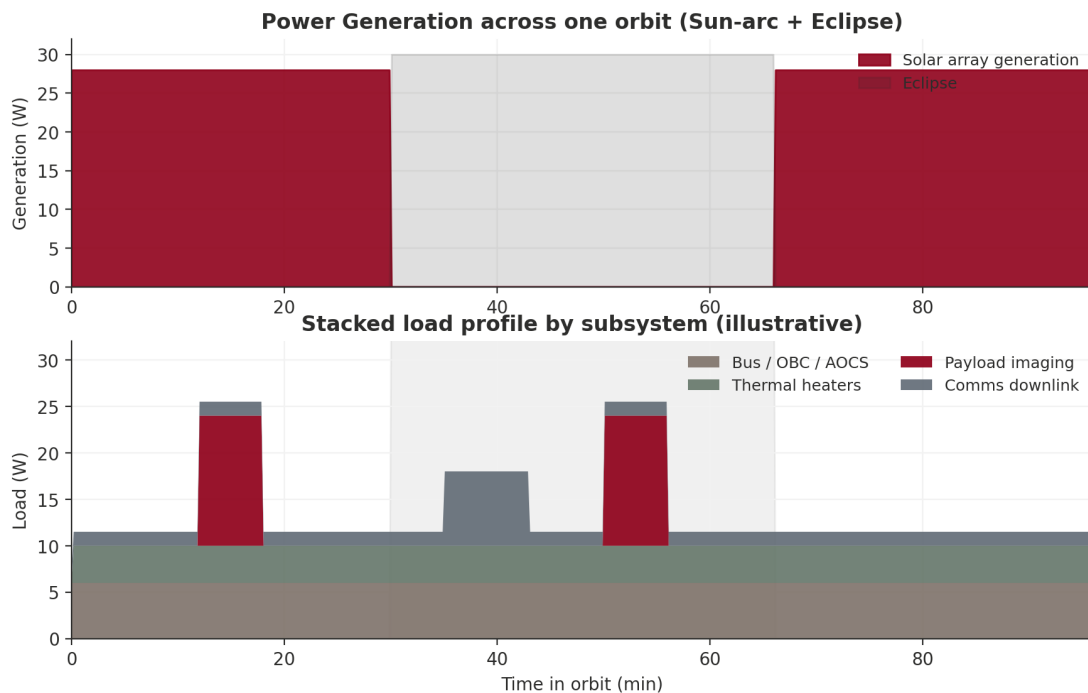
EXPECTED READING

Expected reading SMAD4 Ch. 9 (orbit selection), Ch. 11 (power), Ch. 12 (thermal), Ch. 13 (structures). All referenced sections cached on Brightspace.

Day 3 — Subsystem design loop (8 h)

SESSION 3.1 — POWER & EPS (2 H)

- Power-budget structure: bus / housekeeping / payload / comms / thermal / margin.
- Solar array sizing (body-mounted vs deployable) at end-of-life.
- Battery sizing — DoD vs cycle life trade.
- EPS architecture: MPPT, PCDU, single-bus vs dual-bus.



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Figure 2.6 — Generation (top) and stacked load (bottom) over one orbit. Eclipse drives battery sizing; comms and payload pulses drive PCDU peak rating.

- **Exercise:** size SA + battery in SpaceCDF.
- **Worksheet:** EPS sizing sheet, end-of-life worst-case.

SESSION 3.2 — STRUCTURE & MECHANISMS (2 H)

- Cal Poly CDS Rev 14 — CubeSat structural envelope.
- Launch loads: quasi-static, sine-vibration, random vibration, shock.
- Mass-budget convergence under ECSS margin policy.

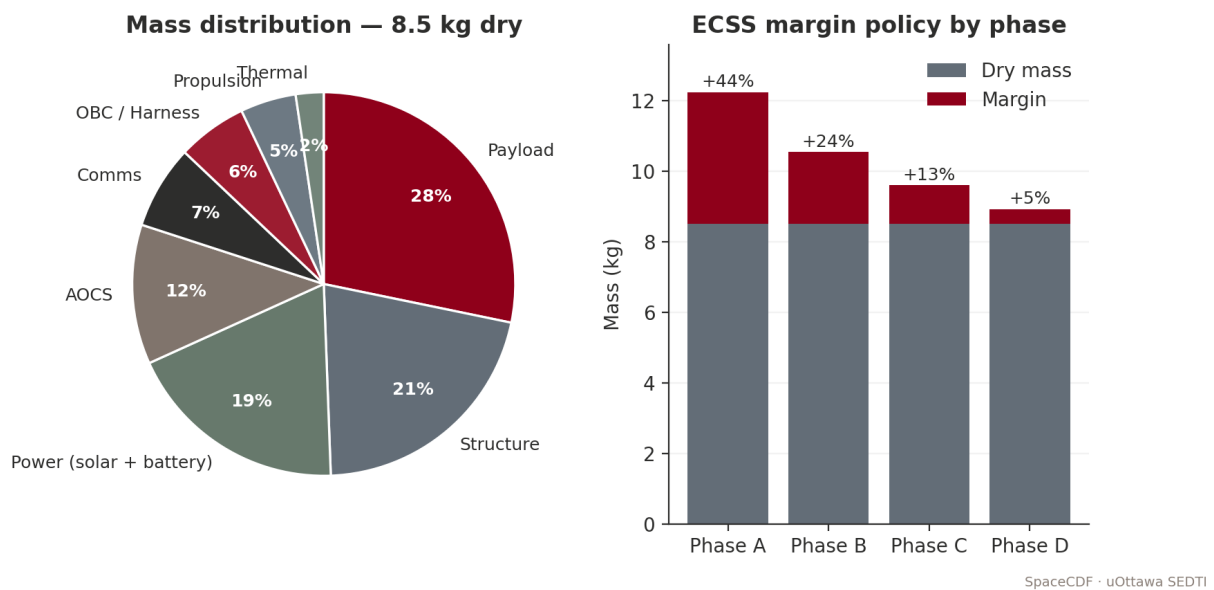


Figure 2.7 — Mass distribution and the ECSS margin schedule. Margin starts at +44 % at Phase A and tightens to +5 % at Phase D.

- **Exercise:** size structure in SpaceCDF.
- **Worksheet:** mass budget with margin column.

SESSION 3.3 — THERMAL (2 H)

- Radiative equilibrium first-order analysis.
- Hot-case / cold-case envelope.
- β -angle and eclipse — see Figure 2.8.
- Passive vs active thermal control; MLI; heaters; heat pipes.

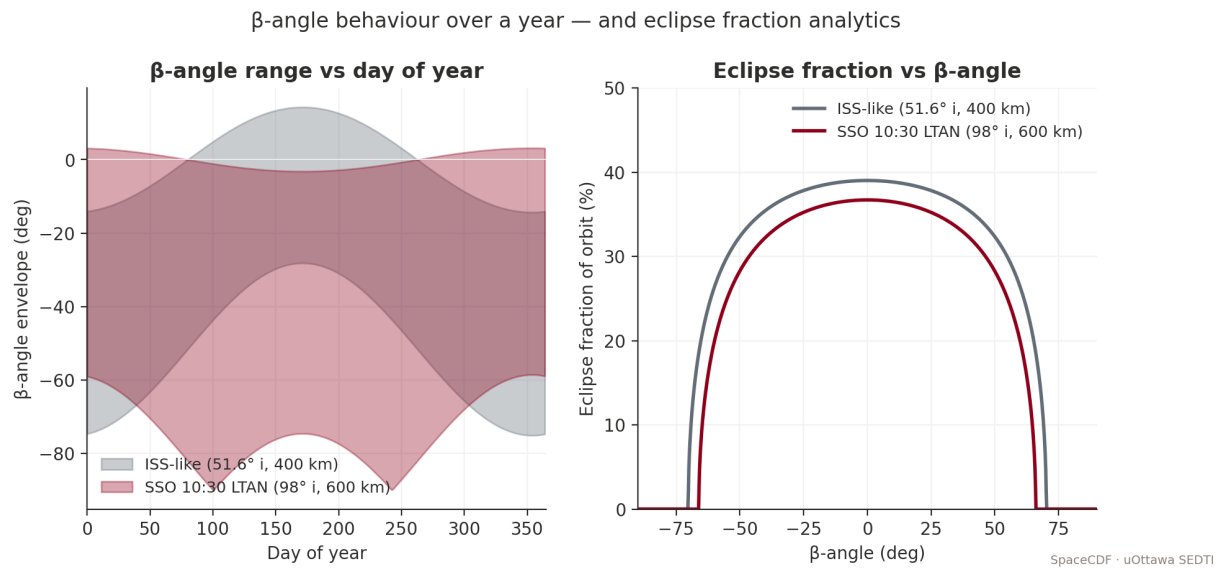


Figure 2.8 — β -angle range vs day of year (left) and eclipse fraction analytical curves (right) for an ISS-like and a 600 km SSO.

- **Exercise:** thermal envelope analysis in SpaceCDF.
- **Worksheet:** thermal budget with heater duty.

SESSION 3.4 — COMMUNICATIONS (2 H)

- Link-budget basics: EIRP, FSPL, G/T, C/N_0 vs E_b/N_0 .
- Frequency band selection — UHF / S / X / Ka.
- Modulation and coding: QPSK + LDPC (DVB-S2 family) for ground.
- Doppler and Doppler-rate budget for a polar pass.
- Antenna patterns and pointing.

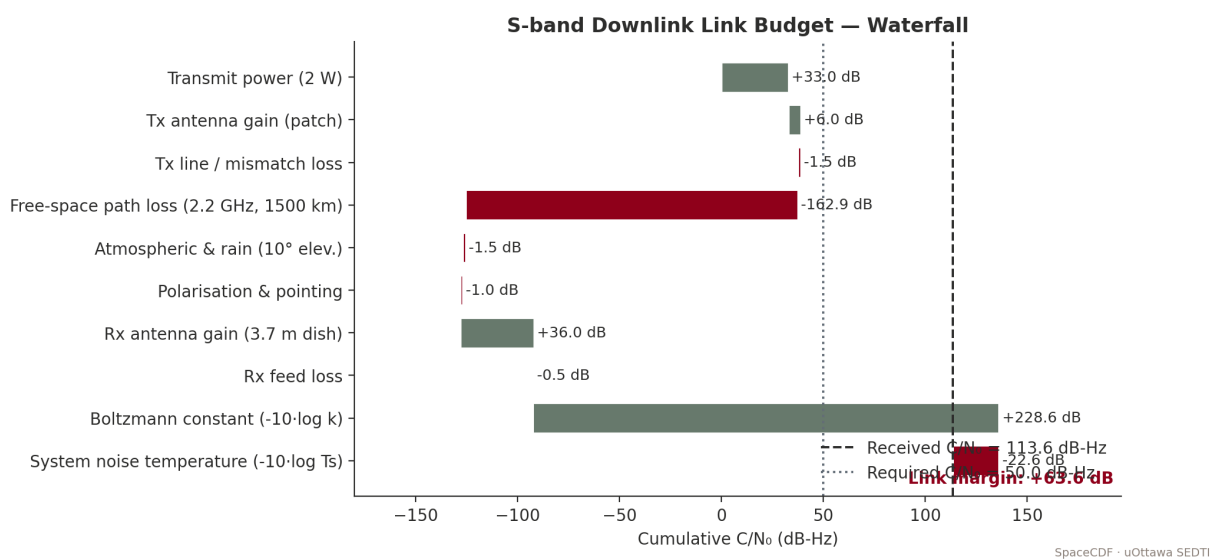


Figure 2.9 — A canonical link-budget waterfall. Required C/N_0 and link margin always sit at the bottom-right.

- **Exercise:** close link budget in SpaceCDF.

- **Worksheet:** link-budget calculator with margin.

EXPECTED READING

Expected reading SMAD4 Ch. 17 (V&V); ECSS-E-ST-10-02C; NASA NPR 8715.3 (system safety — primer only).

Day 4 — Equipment, V&V, risk, cost (8 h)

SESSION 4.1 — EQUIPMENT SELECTION (2 H)

- Catalogue vs custom design.
- TRL gating decisions.

Technology Readiness Level (TRL) Ladder

TRL 9	Flight-proven	TRL 1-3 Discussed in Course Plan §2 — research / IRAD context
TRL 8	Qualified system	
TRL 7	Operational demo	
TRL 6	Relevant-environment demo	TRL 4-6 CubeSat platforms typically buy at TRL ≥ 6 (commercial COTS)
TRL 5	Relevant-environment validation	
TRL 4	Lab validation	TRL 7 First in-orbit demonstration — relevant for new instruments
TRL 3	Proof of concept	
TRL 2	Concept formulated	TRL 8-9 Operational missions — flight heritage drives selection in CDF Day 3
TRL 1	Basic principles	

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Figure 2.10 — TRL ladder. Most CubeSat platform decisions land at TRL 7 – 9; novel instruments at TRL 4 – 6 require explicit risk-buy-down planning.

- **Exercise:** complete equipment selection in SpaceCDF.
- **Worksheet:** equipment-compatibility checklist.

SESSION 4.2 — VERIFICATION & VALIDATION PLANNING (2 H)

- NASA SEH Processes 7 – 8.
- ECSS-E-ST-10-02C — verification methods (A · T · R · I).
- Verification-matrix structure.

- Test levels — unit → subsystem → system → acceptance → qualification.
- Environmental testing — vibration, thermal-vacuum, EMC.
- **Exercise:** review compliance matrix in SpaceCDF.
- **Worksheet:** V&V matrix for 10 key requirements.

SESSION 4.3 — RISK MANAGEMENT (2 H)

- ECSS-M-ST-80C — risk-management process.
- Risk identification, assessment (5×5 matrix), mitigation.
- FMECA methodology (ECSS-Q-ST-30-02C).
- Single-point failure analysis.
- Technical Performance Measures (TPMs).

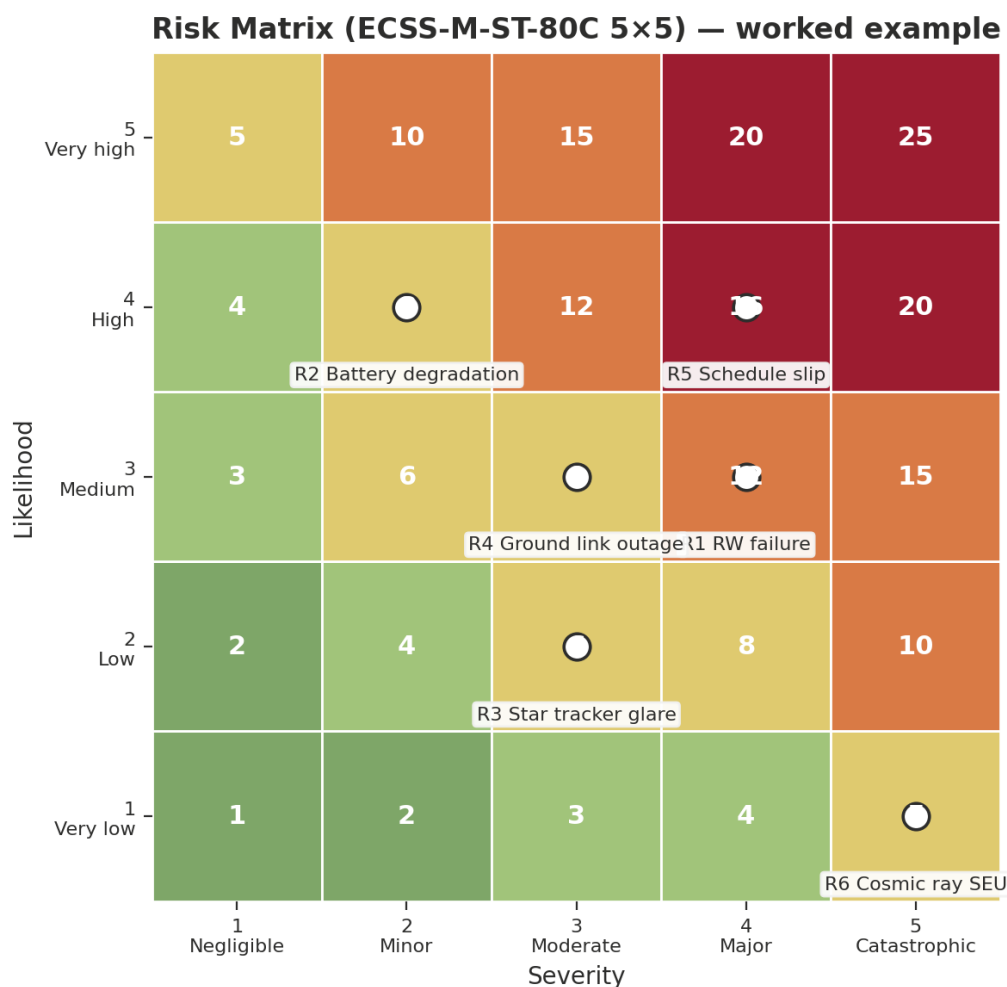


Figure 2.11 — The risk matrix in use. Each plotted point is a real risk on a representative mission; positions on the matrix drive mitigation priority.

- **Exercise:** review risk scores and reliability in SpaceCDF.
- **Worksheet:** risk register for top 5 risks.

SESSION 4.4 — COST & SCHEDULE (2 H)

- Cost estimation: parametric (SSCM, COMPACT), analogy, bottom-up.
- WBS structure (NPR 7120.5).
- CubeSat cost drivers and cost fractions.
- Schedule estimation and critical path; learning-curve effects for constellations.
- **Exercise:** review cost breakdown and run optimiser.
- **Worksheet:** cost estimate by WBS element.

EXPECTED READING

Expected reading ECSS-M-ST-10C §6 (review gates) — <https://ecss.nl/>. ITU Radio Regulations Article 21 — <https://www.itu.int/pub/R-REG-RR>.

Day 5 — Design review & regulatory (8 h)

SESSION 5.1 — GATE REVIEW PREPARATION (2 H)

- ECSS-M-ST-10C — review-gate structure.
- MCR / SRR / PDR / CDR exit criteria.
- Design-review presentation structure: 6-slide standard pack.
- Action-item management.
- **Exercise:** run gate review in SpaceCDF; resolve action items.
- **Worksheet:** PDR presentation outline.

SESSION 5.2 — REGULATORY & LICENSING (2 H)

- Frequency licensing — ITU, IARU, national authority.
- ITU filing process (API → coordination → notification).
- Canadian RSSSA for remote sensing.
- Export control: ITAR / EAR / Canadian CGP.
- UN Registration Convention (COPUOS).
- Space-debris regulations (25-yr / 5-yr rules).
- **Exercise:** generate regulatory filings in SpaceCDF.
- **Worksheet:** licensing decision tree.

SESSION 5.3 — LAUNCH INTEGRATION (2 H)

- Launch providers and pricing landscape.
- Deployer standards (ISIPOD, EXOpod, CSD).
- Launch ICD requirements (mechanical, electrical, environmental).
- Separation switches and inhibits.
- Environmental-test specification derivation.
- **Exercise:** select launch provider, review ICD requirements.

- **Worksheet:** launch ICD compliance checklist.

SESSION 5.4 — DESIGN OPTIMISATION & FINAL REVIEW (2 H)

- Multi-objective optimisation (Pareto concepts).
- Sensitivity analysis (Morris screening).
- Design iteration and convergence.
- Final design-review presentation.
- Lessons-learned capture.
- **Exercise:** run optimiser, review Pareto front, select final design.
- **Final Exercise:** each team presents its complete mission design.

Assessment

- **Day 1 – 4:** worksheets completed in class (formative).
- **Day 5:** team presentation of complete mission design (summative).
- **Post-course:** continued access to the SpaceCDF tool and the Facilitator's Book for self-study.

BLOOM-LEVEL OUTCOME MATRIX

Outcome	Bloom level	Where earned
Recall the 17 SEH processes and gate sequence	Remember	Sessions 1.1, 5.1
Explain the System-V and where it fails	Understand	Sessions 1.1, 4.2
Apply SMART criteria to derive requirements	Apply	Session 2.1
Apply the N ² matrix to a real CubeSat	Apply	Session 2.3
Analyse a link budget and find the dominant loss	Analyse	Session 3.4
Analyse a risk register and find single points of failure	Analyse	Session 4.3
Evaluate trade-off options and defend a recommendation	Evaluate	Sessions 1.3, 5.4
Create a complete preliminary CubeSat design	Create	Day 5 final review

EQUIPMENT & SOFTWARE REGISTER

Tool	Purpose	Where in course
SpaceCDF (web)	Concurrent design and review	All days
Brightspace	Reading distribution, submissions	All days
Python 3.11+ (notebooks)	Optional analytical exercises	Days 3, 4
Open-source plotting (matplotlib, plotly)	Visualisations	Days 3, 4
Office suite	PDR slide pack	Days 1, 5

PRE-REQUISITE KNOWLEDGE AUDIT (SELF-CHECK)

Topic	Required level	Where to refresh
Newtonian mechanics	Year 1 engineering	Hibbeler / Meriam
Vector and matrix algebra	Year 2 engineering	Strang
Basic electromagnetism	Year 1 physics	Griffiths Ch. 1 – 3
Statistics & probability	Year 2 engineering	Montgomery & Runger
Programming basics	Comfortable in any language	—

RACI FOR DELIVERY

Activity	Instructor	Co-instructor	TA / facilitator	Cohort
Lecture delivery	R	A	C	I
CDF session facilitation	A	R	R	C
Tool support	I	I	R	A
Worksheet review	C	R	A	I
Gate review	R	A	C	C
Final review judging	R	R	C	I

R = Responsible, A = Accountable, C = Consulted, I = Informed.

Per-position deep dives (appendices)

The Facilitator's Book carries full per-position appendices (systems engineer, mission analyst, payload, power, AOCS, thermal, structures, propulsion, comms, OBDH, software, operations, ground segment). The summary list of positions and their responsibilities is below; full appendices live in the Facilitator's Book.

Position	Owns (parameters)	Drives (sessions)
Systems Engineer	Mass margin, power margin, system cost, TRL, health score	All
Mission Analyst	Orbit, ground track, contact time, eclipse, β	1.4, 3.1 – 3.3
Payload Engineer	Aperture, GSD, data rate, FOV	1.2, 3.4
Power Engineer	Solar-array area, battery capacity, EPS topology	3.1
AOCS Engineer	Pointing budget, RW capacity, sensor selection	3.2, 4.3
Thermal Engineer	Hot/cold case, radiator area, heater duty	3.3
Structures Engineer	Mass-budget closure, deployers, mechanisms	3.2
Propulsion Engineer	Δv budget, thruster sizing, propellant mass	(when needed)
Comms Engineer	Link budget, antenna pattern, modulation	3.4
OBDH / Software	OBC, FSW architecture, FDIR, telemetry list	4.1
Operations	Pass plan, ConOps, anomaly tree	1.4, 5.1
Ground Segment	Ground stations, MCS, MPS, archive	1.4, 5.1
Cost / Schedule	WBS, CER, critical path	4.4

Reference list (live links)

- **NASA SEH (SP-2016-6105 Rev 2)** — Systems Engineering Handbook. <https://www.nasa.gov/reference/systems-engineering-handbook/>
- **NPR 7123.1D** — <https://nodis3.gsfc.nasa.gov/>
- **NPR 7120.5F** — <https://nodis3.gsfc.nasa.gov/>
- **NASA CubeSat 101** — https://www.nasa.gov/wp-content/uploads/2017/03/nasa_c-sli_cubesat_101_508.pdf
- **ECSS standards (all)** — <https://ecss.nl/>
- **Cal Poly CDS Rev 14** — <https://www.cubesat.org/cds-announcement>
- **CCSDS PUS** — <https://public.ccsds.org/Pubs/660x0g3.pdf>
- **ITU Radio Regulations** — <https://www.itu.int/pub/R-REG-RR>
- **ISED CPC-2-6-02** — <https://ised-isde.canada.ca/site/spectrum-management-telecommunications/en>
- **Remote Sensing Space Systems Act** — <https://laws-lois.justice.gc.ca/eng/acts/R-5.4/>
- **IADC debris guidelines** — <https://www.iadc-home.org/>
- **UN COPUOS Registration Convention** — <https://www.unoosa.org/>
- **Wertz, Everett & Puschell** — Space Mission Engineering: The New SMAD (2011).

- **Larson & Wertz** — Space Mission Analysis and Design, 4th ed.
- **Sutton & Biblarz** — Rocket Propulsion Elements, 9th ed.
- **Markley & Crassidis** — Fundamentals of Spacecraft Attitude Determination and Control.
- **Pratt, Bostian & Allnutt** — Satellite Communications, 3rd ed.
- **Fortescue, Stark & Swinerd** — Spacecraft Systems Engineering, 4th ed.